

THE COMPLETE GUIDE

BENT TOY RACK

Bent Toy Rack

7 circuit-bent toys (sampler included) + studio
+ auto-composer, all covered!





★ Every button explained ★ 50 glitches ★ Killer combos ★ Crash strategy ★ Recommended seeds ★

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Introduction: what is a circuit-bent toy?

Welcome to the world of broken sound!

This is a rack of children's toy instruments picked up at a junk market and modded to the extreme: circuit-bent toys!

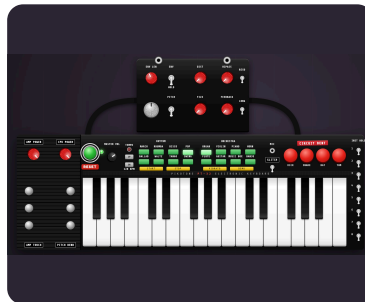
Deliberately rewiring a toy's circuits and adding buttons and knobs to get sounds its makers never imagined is called **circuit bending**. This app builds 7 of those modded toys (sampler included) entirely inside your computer. Unlike the real thing, you can't break them and you can't get a shock!



1. BLIPPY BOOK 30

It talks! It sings! It gets it wrong! The modded toy PC

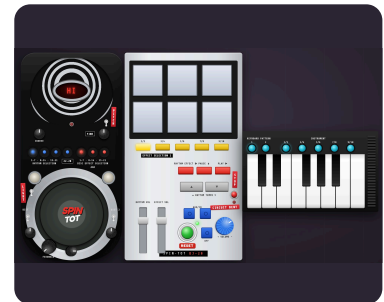
A children's computer with a screen in the lid and 26 letter keys plus do-re-mi number keys. Its 8 modes read letters, words, play melodies, piano, drums, sound effects and a quiz. Add GLITCH buttons, LOOP, STRETCH and DIST, and you get a modded machine whose voice falls apart!



2. PIKOTONE PT-32

Lower the voltage and both sound and soul start to wobble

A 32-key mini keyboard with a rhythm machine and 8 sounds (ORCHESTRA). A separate effects unit sits on top, and on the left we added voltage-starve knobs (deliberately cutting the power) and touch points. A proper modded machine!



3. SPIN-TOT DJ-28

Scratch it! Stop it! Shade it!

A children's DJ set: scratch the disc, 6 sound-effect pads, a 13-key keyboard and 28 rhythms (7 of them hidden). We added coarse/fine pitch, a tape-stop STOP, a light sensor, two DISTs and FEEDBACK. A modded machine!



4. VROOMBOX VR-5

Play the engine's explosions as a scale!

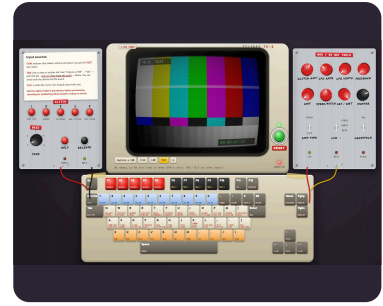
A children's driving dashboard (our own design). The engine sound is made of "ignition grains": skip some with the mod panel and they become rhythms, and TUNE turns them into a scale. Its pride: you can play the engine itself from MIDI or the sequencer!



5. TYPOTRON TT-109

Play the keyboard as an instrument!

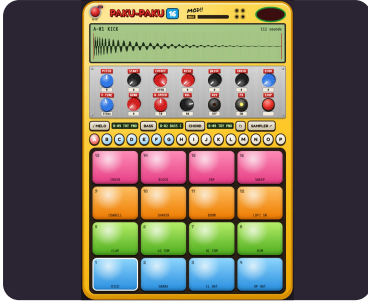
A PC keyboard turned into an instrument (our own design). Every key has a job: the letter rows are scales, the numpad is drums, and the F keys pick waveforms, scales and "keyboard faults". With 12 extra knobs on top, it's a modded machine!



6. TELEKEY TK-6

Wreck the picture and the sound together!

A modded keyboard with a CRT monitor (our own design). It captures picture and sound from a webcam, another tab or a video file and wrecks them with keys. The small text under each key is "picture effect / sound effect". One-shot GLITCH x BASE glitches, 8 knobs and heat runaways: a video machine!



7. PAKU-PAKU 16

**Eat sounds, chop them,
arrange them!**

A sampler made by modding a children's recording toy (with a crocodile-mouth speaker). Fill 16 pads × 16 banks with recorded voices, loaded sounds and the 32 built-in ones, then hit them. In the rack and studio this small version plays, records and auto-composes; sound building (recording, chopping, effects, bending, patterns) happens on its own page (see "The sampler").

Uncle
Mod

Each one breaks in its own way: "reads words wrong", "wobbles from too little power", "plays scales with an engine"... Learning each toy's quirks one at a time is the fastest way to get good!

★ Key point!

Switch toys with the tabs at the top (1–7). Hidden toys keep playing, so you can layer them. **STUDIO** lets you line up your favourite toys and make songs together.



Just touch it!

Your first 3 minutes

No power button needed! Touch anywhere on the screen (or press a key) and the toy on screen switches on. Switch to another toy with the tabs and it switches on by itself too.

Toy	Where RESET (reboot) is
1. BLIPPY BOOK 30	The big green button at the top left of the body (the "REBOOT" tag). Press to reboot
2. PIKOTONE PT- 32	The big green button on the left of the control panel (the "RESET" tag)
3. SPIN-TOT DJ- 28	The big green button at the bottom right (the "RESET" tag)
4. VROOMBOX VR-5	The ignition key by the "POWER" tag on the left (the power is always on; hold a click to crank the engine)
5. TYPOTRON TT- 109	The big green button at the top left of the keyboard (the "RESET" tag)
6. TELEKEY TK-6	The big green button at the bottom right of the monitor (the "RESET" tag)
7. PAKU-PAKU 16	The red button at the top left ("RESET": press it and it goes "chomp, chomp" and reboots)

★ Key point!

- The power is always on. The old POWER button is now **RESET** (press to reboot). If something goes wrong, try pressing it.
- If it stops from RESET or running wild, it starts again right away.
- Browsers stay silent until you first interact, but your first touch also gets the sound ready.

Your first 3 minutes

1 **Touch the screen:** anywhere is fine. 1. BLIPPY says "HELLO".

2 **Play any key:** A–Z on your PC keyboard play directly.

3 Play while holding a glitch: on the toy PC the , . / ; : keys are GLITCH 1–5. It breaks only while held!

4 AUTO COMPOSE: press the red button on the green board to the right and a song for that toy is written and starts playing.

Uncle
Mod

Only the VROOMBOX key is different. The power is always on, but while you turn the key the starter cranks and the engine fires up.



Every part explained

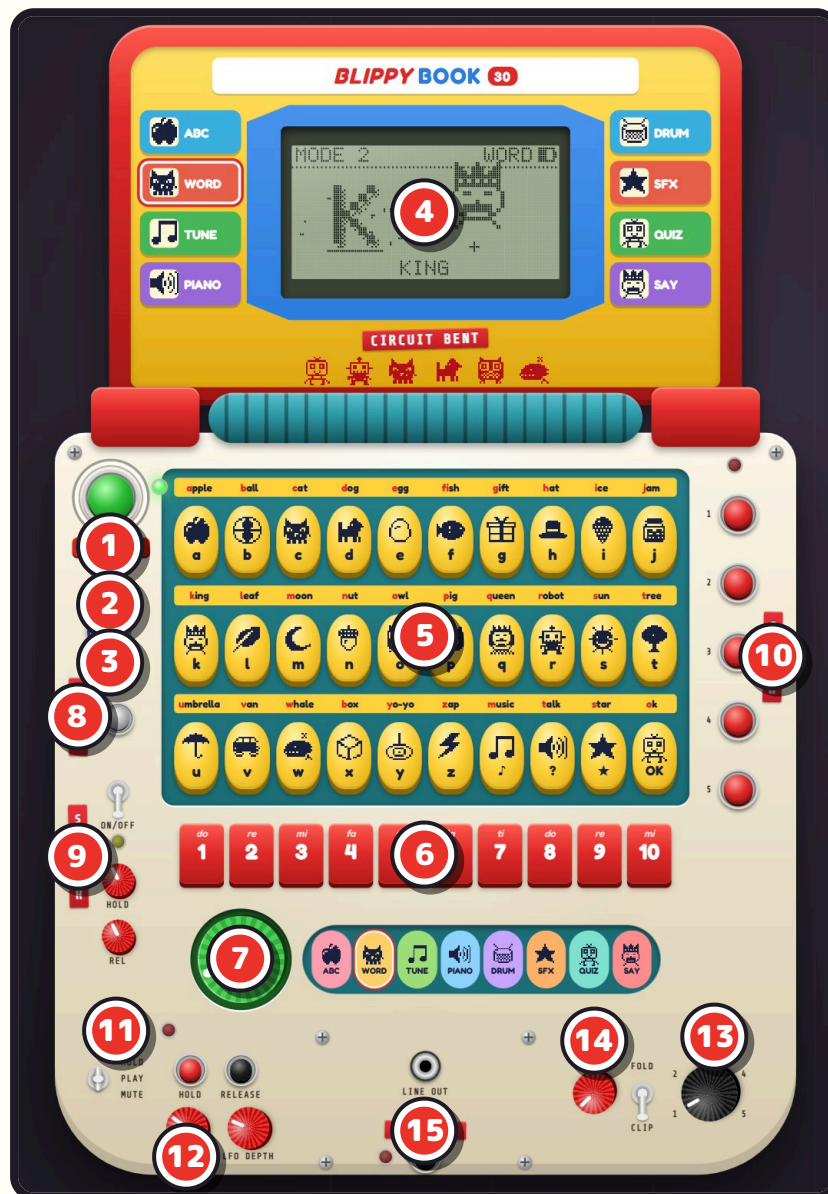
Every button, knob and switch, illustrated!

1

BLIPPY BOOK 30

It talks! It sings! It gets it wrong! The modded toy PC

A children's computer with a screen in the lid and 26 letter keys plus do-re-mi number keys. Its 8 modes read letters, words, play melodies, piano, drums, sound effects and a quiz. Add GLITCH buttons, LOOP, STRETCH and DIST, and you get a modded machine whose voice falls apart!



BLIPPY BOOK 30 (switched on)

1 REBOOT

Reboot. After the startup sound it says "HELLO" and shows the mode screen. Keys do nothing until it finishes talking. The power is always on (it switches on the first time you touch the screen).

2 OFF

Switches the power off. Since it is always on, it comes right back on (reboot).

3 VOLUME

Volume knob.

5 Letter keys (A-Z) and ♪ ? ★ OK

Depending on the mode: letter names, words, melodies, piano, drums, effects or quiz answers. One sound at a time (a new one cuts the old).

💡 A-Z on your PC keyboard are the letter keys. - ^ @ [are ♪ ? ★ OK

7 MODE dial

8 modes: ABC / WORD / TUNE / PIANO / DRUM / SFX / QUIZ / SAY. The picture buttons beside the screen select them too.

💡 PC ← →

9 STRETCH (switch, HOLD, REL)

While on, the sound alternates between "playing on" and "repeating a 25 ms grain". HOLD sets the repeat length (15–900 ms), REL the play length (8–400 ms).

💡 PC Backspace toggles it. Speech goes jerky and robotic

11 LOOP (switch, HOLD, RELEASE)

HOLD grabs and repeats 50 ms of what is playing. RELEASE lets go. Switch: up = grab automatically, middle = normal, down = MUTE (no sound, and it won't even power on!).

💡 PC Space = HOLD, Enter = RELEASE, Tab = switch

13 BASE

Glitch family (1 ADDR, 2 DATA, 3 CLOCK, 4 BEEP, 5 VOICE).

💡 PC ↑ ↓

4 Screen

Shows the letter, character and mode you pressed. During glitches the picture breaks too, and VOICE glitches turn it into a different character.

6 Do-re-mi number keys

10 notes from C to the E above. The same in every mode.

💡 PC keys 1–0

8 RESET

Stops the CPU and leaves it switched off (heat goes back to 0). Press POWER again to start up.

💡 PC Esc. The last resort when it gets too hot

10 GLITCH 1–5

Glitches that work while held. 25 kinds combined with the BASE position (→ chapter 5). Press several at once.

💡 PC , . / ; : keys. Holding them builds heat

12 LFO RATE / DEPTH

Wobbles the pitch only while looping (rate 0.1–25.6 Hz, depth ±1.2 octaves).

💡 Combine with LOOP HOLD. Does nothing on its own

14 DIST (knob, CLIP / FOLD)

Distortion. CLIP squashes, FOLD folds and quantises for a rough sound.

15

MY VOICE

Press to arm → hold a key to record from the mic (up to 4 s). That key becomes your voice. A short press erases it.

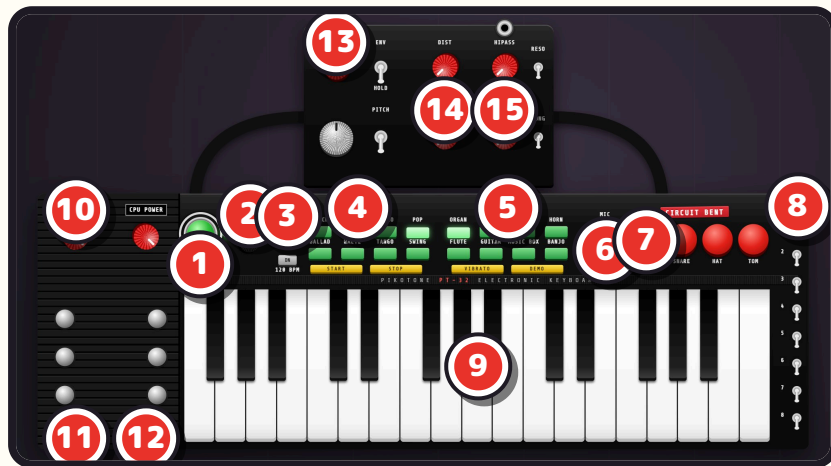


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2 PIKOTONE PT-32

Lower the voltage and both sound and soul start to wobble

A 32-key mini keyboard with a rhythm machine and 8 sounds (ORCHESTRA). A separate effects unit sits on top, and on the left we added voltage-starve knobs (deliberately cutting the power) and touch points. A proper modded machine!



PIKOTONE PT-32 (switched on)

1 RESET

Press to reboot.

3 TEMPO (▲▼)

Rhythm tempo (60–200 BPM).

2 MASTER VOL

Volume.

4 RHYTHM + START / STOP

8 rhythms: MARCH, RHUMBA, DISCO, POP, BALLAD, WALTZ, TANGO, SWING. START plays, STOP stops.

💡 PC ↑ ↓ to choose, Space for START / STOP

5 ORCHESTRA + VIBRATO · DEMO

8 key sounds (ORGAN, VIOLIN, PIANO, HORN, FLUTE, GUITAR, MUSIC BOX, BANJO). VIBRATO wobbles, DEMO plays the demo song.

💡 PC ← → for sounds, Enter for DEMO

7 Red pads

Kick, snare, hat, tom.

💡 PC - ^ @ [

6 GLITCH switch

Turn on to break the rhythm: every snare squeals, sometimes a spot repeats, and the kick pitch wanders (pads too).

💡 PC Backspace

8 INST HOLD 1–8

A mod that holds down the instrument-select lines. The sounds you turn on blend together.

9 Keys (F3–C6)

32 keys. Polyphonic.

💡 **PC Z-/ (white keys), A··· (black keys), Q–P (upper white keys), 2–9 (black keys)**

11 AMP TOUCH 1–3

Only while your finger is on a silver dot: 1 = hum, 2 = distortion, 3 = squealing oscillation. Move up/down while pressing to change the strength.

13 Effects unit: ENV LEN, ENV/HOLD

Note length. On HOLD, notes keep ringing.

💡 **PC Tab for ENV / HOLD**

15 FEEDBACK (LONG)· PITCH

FEEDBACK = the sound loops back on itself (SHORT = 4 ms buzz, LONG = 280 ms echo). PITCH = with the switch on, shifts everything ± 1 octave.

10 AMP POWER / CPU POWER

Voltage starve. Lowering AMP makes the sound quieter, choppy and shaky. Lowering CPU makes everything slower, lower and wobblier; go further and sounds mutate, sometimes stop, and even the notes you play drift (the rhythm too).

💡 **The heart of this machine! Lower it slowly**

12 PITCH BEND 1–3

Bends the pitch only while your finger is on the silver dot. Move up/down while pressing to change the strength.

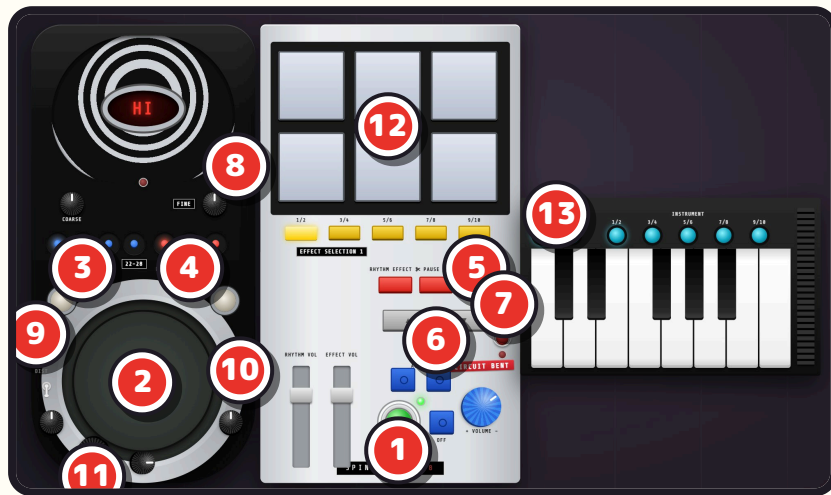
14 DIST · FIZZ · HIPASS (RESO)

DIST = distortion, FIZZ = fizzy noise, HIPASS = cuts the lows (RESO adds a twangy resonance).

3

SPIN-TOT DJ-28 *Scratch it! Stop it! Shade it!*

A children's DJ set: scratch the disc, 6 sound-effect pads, a 13-key keyboard and 28 rhythms (7 of them hidden). We added coarse/fine pitch, a tape-stop STOP, a light sensor, two DISTs and FEEDBACK. A modded machine!



SPIN-TOT DJ-28 (switched on)

1

RESET / OFF

Press green to reboot. The blue OFF also switches it off, and it comes right back on.

2

Disc

Grab and spin it with the mouse to scratch. It scratches the DISC EFFECT sound.

💡 PC . / spin forward / backward (while held)

3

RHYTHM SELECTION

Choose from 28 rhythms (1-7, 8-14, 15-21 and the hidden 22-28).

💡 PC 1 2 3 4. Each press moves on

4

DISC EFFECT SELECTION

Choose the sound the disc scratches (21 kinds).

💡 PC 5 6 7

5

PLAY / PAUSE / RHYTHM EFFECT

Plays / stops the rhythm. RHYTHM EFFECT chops it up while held.

💡 PC Space for PLAY / PAUSE, Enter for EFFECT

6

RHYTHM TEMPO

Rhythm tempo up / down in steps of 4.

💡 PC ↑ ↓

7 STOP (stop moving)

While held, everything slows down and drops in pitch like a stopping tape. Let go to recover.

💡 PC Backspace

9 Light sensor, LIGHT

With LIGHT on, bringing the mouse near the round lenses shades them and lowers the pitch. Pressing on them makes it pitch-dark.

💡 PC L for the switch

11 FEEDBACK (RHY / DISC / ALL)

Feeds the sound back into itself. Choose what to feed back: RHYTHM, DISC or ALL.

13 Keys, KEYBOARD PATTERN, INSTRUMENT

13 keys from C to C. INSTRUMENT picks one of 10 sounds, PATTERN 2 gives arpeggios.

💡 PC A W S E D F T G Y H U J K; ← → for sounds, Tab for pattern

8 PITCH (switch, COARSE, FINE)

With the switch on, shifts the overall pitch. COARSE = ± 12 semitones, FINE = ± 1 semitone.

💡 PC P for the switch

10 DIST · DIST 2

DIST = lopsided distortion, DIST 2 = rectified, rough distortion with an octave up mixed in. Each has on/off and a knob.

12 SOUND EFFECT pads

6 sound effects. EFFECT SELECTION changes the set.

💡 PC Z X C V B N; sets are 8 9 0 - ^

4 VROOMBOX VR-5

Play the engine's explosions as a scale!

A children's driving dashboard (our own design). The engine sound is made of "ignition grains": skip some with the mod panel and they become rhythms, and TUNE turns them into a scale. Its pride: you can play the engine itself from MIDI or the sequencer!



VROOMBOX VR-5 (switched on)

1 POWER (ignition key)

The power is always on (the key sits at ON). While you hold a click the starter cranks and the engine fires up.

💡 **PC Enter (the starter cranks while held)**

3 GEAR

N, 1–5. The higher the gear, the slower the revs catch up. The revs jump the moment you shift.

💡 **PC Z / X**

5 Steering • Horn

Grab the rim to turn it (it springs back). The centre is the horn.

💡 **PC ← →, Space for the horn**

7 TURBO

Turbo while held. Revs shoot up.

💡 **PC T**

2 VOLUME

Volume.

4 ACCEL (pedal)

Throttle while pressed. Revs (= ignition rate = pitch) go up.

💡 **PC ↑ or W**

6 Radio

Presets 1–8 tune the station. During PRESET HIJACK the preset buttons become an engine scale (C to C).

💡 **PC 1–8, R changes station**

8 SIREN / SIGNAL / WIPERS / CRASH!

Siren, ticking indicator, wipers (on/off); CRASH! is a crash sound.

💡 **PC S / V / B / C**

9 FIRING ORDER 1-8 · CAM

Switches for 8 cylinders. Skip ignitions and low revs turn into rhythms. CAM sets the pattern length (3-8).

💡 **The heart of this machine! Skip some, then just a little throttle**

11 TURBO FB

When on, it feeds back the sound one ignition late, making a metallic ring that follows the revs.

13 STARTER LOOP

While held, the starter motor keeps cranking even with the engine running.

💡 **PC Backspace**

10 REDLINE · SPARK · RADIO BLEED

REDLINE = raises the rev limit (at 0 the limiter skips ignitions; higher revs higher), SPARK = a crackle on each ignition, RADIO BLEED = radio and engine multiply together.

12 TUNE · HIJACK · HORN BEND · GRIND

TUNE = revs snap to a scale, HIJACK = play the engine with the radio buttons, HORN BEND = the horn bends, GRIND = grinding on every gear change.

💡 **PC U for TUNE, H for HIJACK**

14 CHASSIS / HAZARD

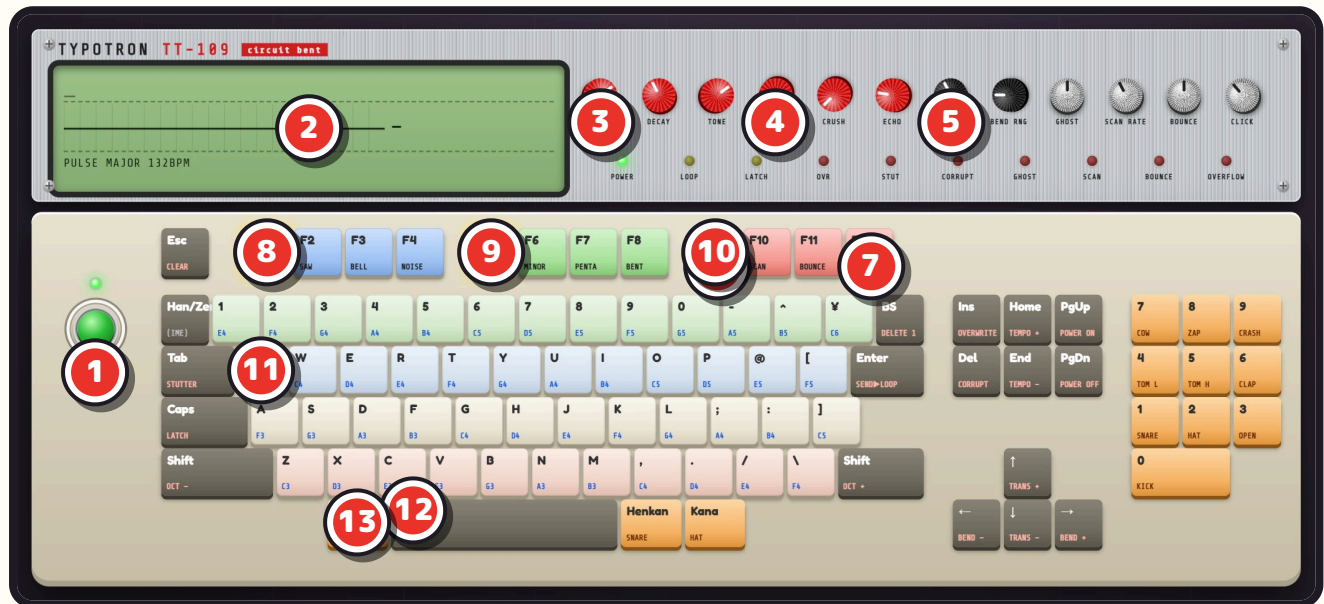
Touch the metal dots: CHASSIS = adds hum and sometimes skips ignitions.

HAZARD = chops the engine finely with more relay clicks.

5 TYPOTRON TT-109

Play the keyboard as an instrument!

A PC keyboard turned into an instrument (our own design). Every key has a job: the letter rows are scales, the numpad is drums, and the F keys pick waveforms, scales and "keyboard faults". With 12 extra knobs on top, it's a modded machine!



TYPOTRON TT-109 (switched on)

1 RESET

Press to reboot.

3 VOLUME • DECAY • TONE

Volume, decay, brightness.

5 TEMPO • BEND RNG

Line-loop speed (60–240 BPM) and pitch-bend range (0–12 semitones).

7 Status LEDs

Which of POWER, LOOP, LATCH, OVR, STUT, CORRUPT, GHOST, SCAN, BOUNCE, OVERFLOW are active.

2 Screen

Shows the "line" of typed notes, waveform, scale, tempo and more.

4 DRIVE • CRUSH • ECHO

Distortion, bit crushing, echo.

6 GHOST • SCAN RATE • BOUNCE • CLICK

Strength of the "keyboard faults" (how often GHOST fires, SCAN speed, BOUNCE count) and the volume of the key clicks.

8 F1-F4: WAVE

4 waveforms: PULSE, SAW, BELL, NOISE.

9 F5-F8: SCALE

MAJOR, MINOR, PENTA, BENT (a deliberately detuned scale).

11 Letter rows: scales

The Z row is lowest; each row up is about a fourth higher. Hold Shift for an octave down (left) / up (right).

13 Numpad, Muhenkan, Henkan, Kana: Drums

KICK · SNARE · HAT · OPEN · TOM L · TOM H
· CLAP · COW · ZAP · CRASH.

10 F9-F12: keyboard faults

GHOST = neighbouring keys sound by themselves, SCAN = reads held keys one by one (an arpeggio), BOUNCE = key bounce repeats, OVERFLOW = the line overflows and runs wild.

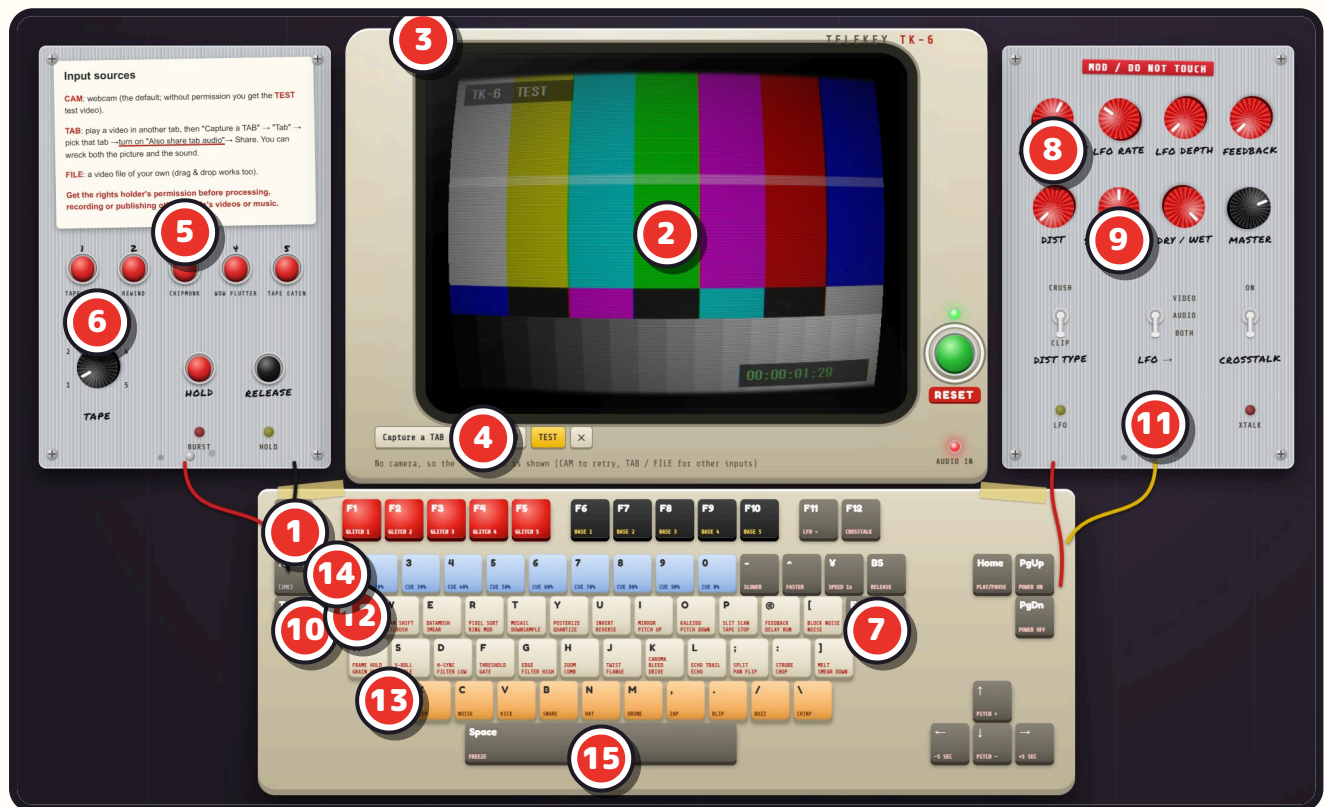
12 Space: SUSTAIN and other function keys

Space = sustain. Enter = loop the typed "line", Tab = STUTTER, Caps = LATCH (keeps ringing), Del = CORRUPT, Ins = overwrite.

6 TELEKEY TK-6

Wreck the picture and the sound together!

A modded keyboard with a CRT monitor (our own design). It captures picture and sound from a webcam, another tab or a video file and wrecks them with keys. The small text under each key is "picture effect / sound effect". One-shot GLITCH × BASE glitches, 8 knobs and heat runaways: a video machine!



TELEKEY TK-6 (switched on)

1 RESET

Press to reboot. The CRT comes back on with a "boom".

2 Monitor

Shows the captured picture wrecked by the active glitches (WebGL). Static when there is no input.

3 REC VIDEO

Records the wrecked picture and sound and saves WebM. Press again to stop and save.

4 Input select

Capture a TAB (picture and sound from another tab), FILE (video file), CAM (webcam), TEST (generated test video), × (disconnect). → chapter 8

5 **GLITCH buttons 1–5**

Fires the "one-shot glitch" set by the current BASE for about a second ($5 \times 5 = 25$ kinds, → chapter 5).

💡 **PC F1–F5**

7 **HOLD / RELEASE**

HOLD = keeps the active glitches going after you let go. RELEASE = stops them all.

💡 **PC Enter / BS**

9 **DIST · SPEED/PITCH · DRY/WET · MASTER**

Distortion, playback speed and pitch, dry/wet mix, master volume.

11 **HEAT LED**

Heat (stress). The hotter it is, the faster it blinks red, the screen shakes and glitches run wild. It never freezes.

💡 → **chapter 7**

13 **Z-[: instrument keys**

Beep, noise, kick, snare, hat, drone, zap, blip, buzzer, chirp (pitched).

15 **Space: FREEZE**

While held, grabs and repeats the current picture and sound.

6 **BASE**

5 levels: TAPE, DIGITAL, SIGNAL, BEEP, MELTDOWN. Changes the one-shot glitch.

💡 **PC F6–F10**

8 **GLITCH AMT · LFO RATE · LFO DEPTH · FEEDBACK**

Glitch strength, wobble speed and depth, and feedback for picture and sound.

10 **DIST TYPE · LFO → · CROSSTALK**

Distortion type (CLIP / CRUSH), LFO target (VIDEO / AUDIO / BOTH), key crosstalk (on: neighbours fire too, or a key does something different each press).

💡 **PC Tab, F11, F12**

12 **Q-[· A-]: glitch keys**

24 glitches that work while held. Picture and sound break together (→ table in chapter 8).

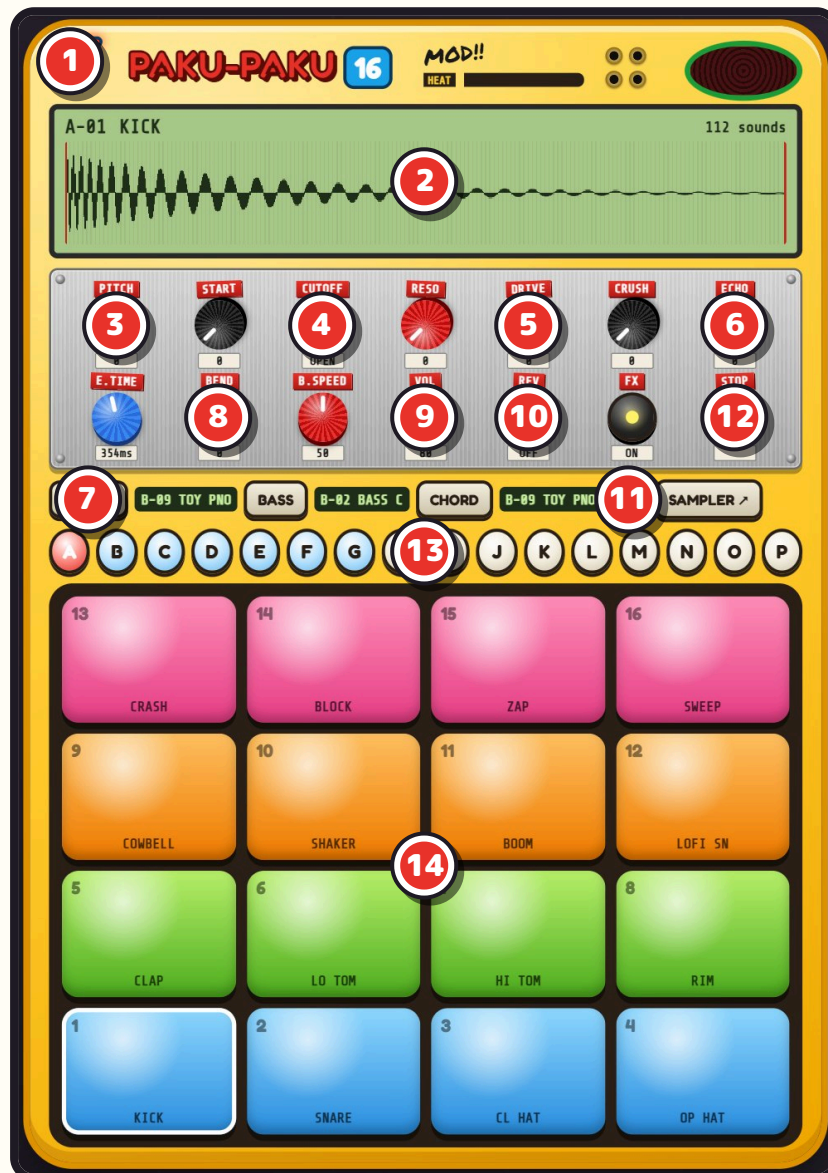
14 **1–0: cue points**

Jump to 10%–90% / 0% of the video. Shift+number stores the current position.

7

PAKU-PAKU 16 Eat sounds, chop them, arrange them!

A sampler made by modding a children's recording toy (with a crocodile-mouth speaker). Fill 16 pads × 16 banks with recorded voices, loaded sounds and the 32 built-in ones, then hit them. In the rack and studio this small version plays, records and auto-composes; sound building (recording, chopping, effects, bending, patterns) happens on its own page (see "The sampler").



PAKU-PAKU 16 (switched on)

1

RESET

Press to reboot. It goes "chomp, chomp".

2

Screen

The selected pad, its sound name and the number of sounds. Below is the waveform: red lines mark the range played (START / END), the bright line is the playhead.

3 PITCH · START

Shift all pads ± 12 semitones / move the start later (the further you turn, the later into the slice).

5 DRIVE · CRUSH

Distort / crush bits and sample rate into grit.

7 🎵 MELO / BASS / CHORD

Use the selected pad for melody, bass or chords. The "🎵", "BASS" and "CHORD" rows in the sequencer play it at different pitches. The auto-composer picks instruments that suit the style here.

💡 Banks C–G hold genre instruments (Japanese, reggae, jazz, dance, pop). Using your own recorded voice as the melody is fun

9 VOL

Volume (it comes after the mods, so distortion doesn't make it too loud).

11 ↺ (reload) · SAMPLER ↗

Reload the sounds made on the sampler page (saved in this browser) / open the sampler page in a new tab.

13 Banks A–P

Switches between sets of 16 pads. Banks with sounds are light blue.

💡 PC []

4 CUTOFF · RESO

Output low-pass. Turn to muffle; RESO adds a peaky resonance.

💡 The auto-composer opens it gradually in build-ups

6 ECHO · E.TIME

Echo amount and spacing (30–750 ms).

8 BEND · B.SPEED

The higher, the more jumper wires on the board and the more playback stutters and crackles (0 = the sampler page setting) / how often it breaks.

💡 MIDI CC1 (modulation) moves it too

10 REV · FX

Reverse all pads / turn the sampler page's effects on or off.

12 STOP

Stops everything that is playing (looping sounds too).

14 16 pads

Bottom left is 1. Press to play. 🎵 and BASS mark the melody and bass pads.

💡 PC Z X C V · A S D F · Q W E R · 1 2 3 4 match the pad layout



Modes and keys: full data

Toy PC: 8 modes × every key

On the toy PC (BLIPPY BOOK 30) the same key makes a different sound in each MODE. Here is every key!

★ Dead keys that make no sound...

There are none! Every key on this toy PC always makes a sound. Even a wrong letter in QUIZ mode says "NO!" or "TRY AGAIN".

Keys	1. ABC	2. WORD	3. TUNE	4. PIANO	5. DRUM	6. SFX	7. QUIZ	8. SAY
A	A	APPLE	Melody A	C3	KICK	LASER	Answer A	A IS FOR APPLE
B	B	BALL	Melody B	D3	SNARE	BOOM	Answer B	B IS FOR BALL
C	C	CAT	Melody C	E3	HAT	COIN	Answer C	C IS FOR CAT
D	D	DOG	Melody D	F3	OPEN HAT	JUMP	Answer D	D IS FOR DOG
E	E	EGG	Melody E	G3	TOM	SIREN	Answer E	E IS FOR EGG
F	F	FISH	Melody F	A3	CLAP	RING	Answer F	F IS FOR FISH
G	G	GIFT	Melody G	B3	COWBELL	ROBOT	Answer G	G IS FOR GIFT
H	H	HAT	Melody H	C4	ZAP	ARPEGGIO	Answer H	H IS FOR HAT
I	I	ICE	Melody I	D4	KICK	FALL	Answer I	I IS FOR ICE
J	J	JAM	Melody J	E4	SNARE	CREATURE	Answer J	J IS FOR JAM
K	K	KING	Melody K	F4	HAT	LASER	Answer K	K IS FOR KING
L	L	LEAF	Melody L	G4	OPEN HAT	BOOM	Answer L	L IS FOR LEAF
M	M	MOON	Melody M	A4	TOM	COIN	Answer M	M IS FOR MOON
N	N	NUT	Melody N	B4	CLAP	JUMP	Answer N	N IS FOR NUT
O	O	OWL	Melody O	C5	COWBELL	SIREN	Answer O	O IS FOR OWL
P	P	PIG	Melody P	D5	ZAP	RING	Answer P	P IS FOR PIG
Q	Q	QUEEN	Melody Q	E5	KICK	ROBOT	Answer Q	Q IS FOR QUEEN
R	R	ROBOT	Melody R	F5	SNARE	ARPEGGIO	Answer R	R IS FOR ROBOT
S	S	SUN	Melody S	G5	HAT	FALL	Answer S	S IS FOR SUN
T	T	TREE	Melody T	A5	OPEN HAT	CREATURE	Answer T	T IS FOR TREE

Keys	1. ABC	2. WORD	3. TUNE	4. PIANO	5. DRUM	6. SFX	7. QUIZ	8. SAY
U	U	UMBRELLA	Melody U	B5	TOM	LASER	Answer U	U IS FOR UMBRELLA
V	V	VAN	Melody V	C6	CLAP	BOOM	Answer V	V IS FOR VAN
W	W	WHALE	Melody W	D6	COWBELL	COIN	Answer W	W IS FOR WHALE
X	X	BOX	Melody X	E6	ZAP	JUMP	Answer X	X IS FOR BOX
Y	Y	YO-YO	Melody Y	F6	KICK	SIREN	Answer Y	Y IS FOR YO-YO
Z	Z	ZAP	Melody Z	G6	SNARE	RING	Answer Z	Z IS FOR ZAP
♪	♪ MUSIC (the mode's melody)	♪ MUSIC (the mode's melody)	♪ MUSIC (the mode's melody)	Chord C	BEAT 1 (a 1-bar rhythm)	♪ MUSIC (the mode's melody)	Repeat the question	♪ MUSIC (the mode's melody)
?	? Says a catchphrase	? Says a catchphrase	? Says a catchphrase	Chord F	BEAT 2 (a 1-bar rhythm)	? Says a catchphrase	Next question	? Says a catchphrase
★	★ Sound effect	★ Sound effect	★ Sound effect	Chord G	BEAT 3 (a 1-bar rhythm)	★ Sound effect	Hint (the answer letter)	★ Sound effect
OK	OK "GOOD JOB"	OK "GOOD JOB"	OK "GOOD JOB"	Chord Am	BEAT 4 (a 1-bar rhythm)	OK "GOOD JOB"	"GOOD JOB"	OK "GOOD JOB"
1-0	In every mode: do-re-mi notes (1=do · 2=re · 3=mi · 4=fa · 5=so · 6=la · 7=ti · 8=do · 9=re · 10=mi)							

QUIZ mode: press the letter of the character on screen for "RIGHT!" → next question. A wrong letter gives "NO!" or "TRY AGAIN". In TUNE mode each letter's "melody" is a different short tune.

⚠ Careful!

Only one sound at a time. A new key stops the previous sound (just like the real toy). Press another key halfway through a long "K IS FOR KING" and it gets cut off.



The glitch encyclopedia

All 50 GLITCH × BASE combinations, with ferocity ratings!

Toy PC: GLITCH 1–5 × BASE 1–5 = 25 kinds

They work only while held. Press as many as you like at once (they stack). Ferocity shows how violently the sound breaks.

BASE 1 ADDR (the read position breaks)

Button	Name	What happens?	Ferocity
GLITCH 1	SKIP	The sound jumps ahead all over. Speech becomes fast and chopped	★★★★★
GLITCH 2	STUTTER	Machine-guns a short slice of where you pressed	★★★★★
GLITCH 3	REWIND	Rewinds a little, then moves on: "Ki, ki, king"	★★★★★
GLITCH 4	STUCK ADDR	Part of the read position gets stuck, so only certain spots repeat	★★★★★
GLITCH 5	REVERSE	Jitters back and forth, with bits of reverse	★★★★★

BASE 2 DATA (the sound data breaks)

Button	Name	What happens?	Ferocity
GLITCH 1	BIT FLIP	Data bits flip and it fizzes and mutates	★★★★★
GLITCH 2	LOW BITS	Fine detail vanishes into gritty low fidelity	★★★★★
GLITCH 3	MSB HIGH	The top bit gets stuck and the sound flips into a blast	★★★★★
GLITCH 4	NIBBLE SWAP	The data flips upside down and the voice becomes a lump of noise	★★★★★
GLITCH 5	DROPOUT	"Normal / frozen / silent" switch rapidly	★★★★★

BASE 3 CLOCK (the speed breaks)

Button	Name	What happens?	Ferocity
GLITCH 1	SLOW	Slower and lower, 0.25–0.66× (fixed when you press)	★★★★★

GLITCH 2	FAST	Faster and higher, 1.5–4× (fixed when you press)	★★★★★
GLITCH 3	WOBBLE	The pitch wobbles like jelly	★★★★★
GLITCH 4	STAIRS	The pitch jumps in steps like a staircase	★★★★★
GLITCH 5	STALL	Stops and lurches forward over and over	★★★★★

BASE 4 BEEP (a buzzer mixes in)

Button	Name	What happens?	Ferocity
GLITCH 1	RUNAWAY	A beeping square wave follows the loudness	★★★★★
GLITCH 2	RING MOD	Metallic ring modulation	★★★★★
GLITCH 3	BEEP ARP	Random-pitched beeps race by every 30 ms	★★★★★
GLITCH 4	NOISE GATE	Becomes noise in the shape of the voice (like whispering)	★★★★★
GLITCH 5	FEEDBACK	A loop where the output changes the speed. Once it runs wild it won't stop	★★★★★

BASE 5 VOICE (turns into another voice)

Button	Name	What happens?	Ferocity
GLITCH 1	MISREAD	Turns into the voice of another key in the same mode (the screen character changes too)	★★★★★
GLITCH 2	SCRAMBLE	Jumps from one key's voice to another	★★★★★
GLITCH 3	GARBAGE	A jumble of fragments from other keys and system sounds (startup sound etc.)	★★★★★
GLITCH 4	MACHINEGUN	Keeps jumping back to where you pressed: ba-ba-ba-ba	★★★★★
GLITCH 5	HICCUP	The startup sound and "RIGHT!" butt in like hiccups	★★★★★

TELEKEY: GLITCH buttons (F1–F5) × BASE (F6–F10) = 25 one-shot glitches

One press wrecks the picture and the sound together for about a second (see "Length"). A one-shot move.

BASE 1 TAPE

Button	Name	What happens?	Length	Ferocity
F1	TAPE STOP	Drops in pitch like a stopping tape Glitches applied: V-ROLL / WOBBLE+MELT / SMEAR DOWN	1 s	★★★★★
F2	REWIND	Reverse spin like rewinding Glitches applied: V-ROLL / WOBBLE+RGB SHIFT / STUTTER	0.8 s	★★★★★
F3	CHIPMUNK	Faster and higher (chipmunk voice) Glitches applied: ZOOM / COMB+CHROMA BLEED / DRIVE	0.8 s	★★★★★
F4	WOW FLUTTER	Tape wobble (wow and flutter) Glitches applied: TWIST / FLANGE+H-SYNC / FILTER LOW	1.2 s	★★★★★
F5	TAPE EATEN	The tape gets chewed and the sound drops out here and there Glitches applied: BLOCK NOISE / NOISE+SCAN SHIFT / BITCRUSH	1 s	★★★★★

BASE 2 DIGITAL

Button	Name	What happens?	Length	Ferocity
F1	BUFFER LOOP	Repeats a short buffer Glitches applied: SPLIT / PAN FLIP+RGB SHIFT / STUTTER	0.9 s	★★★★★
F2	2-BIT	Combines picture and sound glitches in one hit Glitches applied: POSTERIZE / QUANTIZE+THRESHOLD / GATE+SCAN SHIFT / BITCRUSH	0.8 s	★★★★★
F3	LO-FI	Combines picture and sound glitches in one hit Glitches applied: MOSAIC / DOWNSAMPLE+MOSAIC / DOWNSAMPLE	0.9 s	★★★★★
F4	SCRAMBLE	Scatters it into fragments everywhere Glitches applied: DATAMOSH / SMEAR+BLOCK NOISE / NOISE	1 s	★★★★★
F5	DROPOUT	Rapid dropouts Glitches applied: STROBE / CHOP+FRAME HOLD / GRAIN HOLD	0.8 s	★★★★★

BASE 3 SIGNAL

Button	Name	What happens?	Length	Ferocity
F1	RING SWEEP	The ring-modulator frequency keeps rising Glitches applied: RGB SHIFT / STUTTER+PIXEL SORT / RING MOD	1 s	★★★★★
F2	SYNC LOST	Static (noise) on top Glitches applied: H-SYNC / FILTER LOW+V-ROLL / WOBBLE+BLOCK NOISE / NOISE	1 s	★★★★★
F3	HOWL	A feedback squeal mixes in Glitches applied: FEEDBACK / DELAY RUN+ECHO TRAIL / ECHO	1.2 s	★★★★★
F4	FILTER DIVE	The ring-modulator frequency keeps rising Glitches applied: CHROMA BLEED / DRIVE+H-SYNC / FILTER LOW	1 s	★★★★★
F5	STATIC	Static (noise) on top Glitches applied: BLOCK NOISE / NOISE	0.9 s	★★★★★

BASE 4 BEEP

Button	Name	What happens?	Length	Ferocity
F1	ZAP ROLL	A barrage of zaps Glitches applied: STROBE / CHOP	0.9 s	★★★★★
F2	CHORD BLAST	A crashing chord of beeps Glitches applied: INVERT / REVERSE	0.7 s	★★★★★
F3	DRUM ROLL	Drum roll Glitches applied: ZOOM / COMB	1 s	★★★★★
F4	DRONE SWELL	A drone swells up Glitches applied: KALEIDO / PITCH DOWN	1.5 s	★★★★★
F5	CHIRPS	Chirps fly around Glitches applied: TWIST / FLANGE	1 s	★★★★★

BASE 5 MELTDOWN

Button	Name	What happens?	Length	Ferocity
--------	------	---------------	--------	----------

F1	FREEZE DROP	Drops in pitch like a stopping tape Glitches applied: FRAME HOLD / GRAIN HOLD+EDGE / FILTER HIGH	1.2 s	★★★★★
F2	BACKFIRE	Reverse spin like rewinding Glitches applied: INVERT / REVERSE+FEEDBACK / DELAY RUN+ECHO TRAIL / ECHO	1.2 s	★★★★★
F3	DIVE	Nosedive Glitches applied: MELT / SMEAR DOWN+SLIT SCAN / TAPE STOP+H-SYNC / FILTER LOW	1.5 s	★★★★★
F4	SHATTER	Scatters it into fragments everywhere Glitches applied: SLIT SCAN / TAPE STOP+DATAMOSH / SMEAR+SPLIT / PAN FLIP	1.2 s	★★★★★
F5	CHAOS	5 random glitches at once	1.5 s	★★★★★



Killer combos

Learn these combinations!

BLIPPY BOOK 30

Robot reader



1. Set MODE to SAY
2. Turn STRETCH on (HOLD knob fairly high)
3. Press a letter key

👉 "K IS FOR KING" gets stretched in 25 ms grains into a jerky robot voice.

BLIPPY BOOK 30

LOOP HOLD → LFO full → STRETCH



1. Turn STRETCH on first
2. Press a letter key and, while it sounds, LOOP HOLD (Space)
3. Raise LFO DEPTH and LFO RATE → it wails like a siren
4. Release with LOOP RELEASE (Enter) and STRETCH takes over, stuttering forward

👉 The LFO only works while looping, and LOOP beats STRETCH while looping. So you get a two-stage change: "wail → stutter the moment you release".

BLIPPY BOOK 30

The mispronouncing machine



1. Set MODE to WORD
2. Set BASE to 5 (VOICE)
3. Hold GLITCH 1 (MISREAD) and press letter keys

👉 "KING" turns into another word. Even the screen character changes, so it looks wrong too.

BLIPPY BOOK 30

The buggy drum machine



1. Set MODE to DRUM
2. Set BASE to 3 (CLOCK)
3. Hold GLITCH 4 (STAIRS) and tap a rhythm on the letter keys

👉 The kick and snare pitches jump like stairs: a broken rhythm machine.

BLIPPY BOOK 30

Beep runaway

★★★★★

1. Set BASE to 4 (BEEP)
2. Raise DIST and set it to FOLD
3. Press GLITCH 5 (FEEDBACK) and GLITCH 1 (RUNAWAY) together

👉 The output changes the speed, which changes the output again. Heat builds fast, so a runaway is never far away!

TELEKEY TK-6

Grab and let go

★★★★★

1. Press 2–3 glitch keys
2. Then Enter (HOLD)
3. Release the keys and play a rhythm on the instrument keys (Z...)
4. BS (RELEASE) stops everything

👉 HOLD freezes how the picture and sound are broken, leaving both hands free.

TELEKEY TK-6

Crosstalk roulette

★★★★★

1. F12 to turn CROSSTALK on
2. Press the same glitch key again and again

👉 Each press mixes in a neighbour or does something else. You never know what you'll get.

TELEKEY TK-6

MELTDOWN's CHAOS

★★★★★

1. F10 for BASE 5 (MELTDOWN)
2. F5 for CHAOS

👉 A one-shot move with 5 random glitches at once. Heat jumps too.

PIKOTONE PT-32

The dying-battery demo

★★★★★

1. Enter for DEMO
2. Slowly lower CPU POWER
3. Raise FEEDBACK and set it to LONG

👉 The demo and rhythm get slower, lower and wobblier, with echoes piling up. Go too far and it mutates and stops.

SPIN-TOT DJ-28

Tape stop + scratch

★★★★★

1. PLAY the rhythm
2. Press and release Backspace (STOP)
3. Scratch the disc

👉 STOP is a tape stop that drops in pitch. Combined with scratching it sounds like a DJ.

VROOMBOX VR-5

Engine drums



1. Enter to switch on → hold Enter again to start the engine
2. Turn off some FIRING ORDER switches
3. Press the throttle just a little and turn TUNE on

👉 Ignitions are skipped, so low revs become a rhythm. TUNE snaps the revs to a scale.

TYPOTRON TT-109

The broken arpeggio



1. Turn on Caps (LATCH)
2. Press a few keys that make a chord (no need to hold them)
3. F10 for SCAN, F11 for BOUNCE

👉 "SCAN" reads the held keys one by one into an arpeggio, and "BOUNCE" adds key-bounce repeats on top.

Studio

Auto-compose → lock → redo



1. AUTO COMPOSE
2. Press  on a track you like to make it 
3. Auto-compose again

👉 Only locked tracks stay and the rest is new. Keep a good beat and hunt for a melody.



Crash strategy

Build up heat and hang on at the edge!

The toys in this app **never** freeze just because you play them. Instead, there is "heat (stress)"!

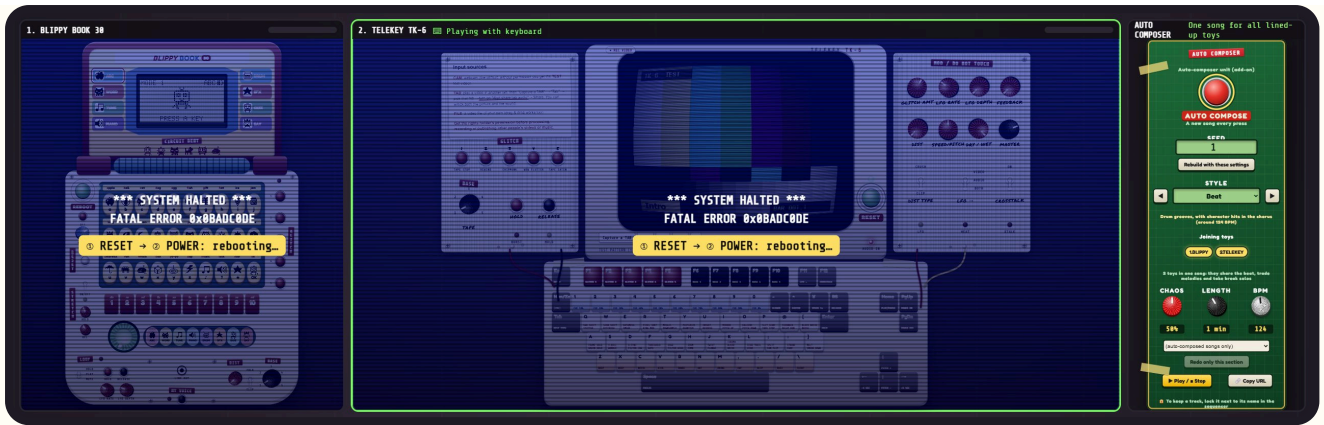
How heat works (toy PC, TELEKEY)

	Toy PC	TELEKEY
Heat rises	The longer you hold GLITCH buttons (much faster with 2 or 3). Higher BASE, with DIST / LOOP / STRETCH on, or hammering keys during glitches	The more glitch keys are active. One-shot glitches, hammering keys, maxed FEEDBACK / DIST / GLITCH AMT and CROSSTALK raise it too
Runaway	Above 40% heat, glitches you aren't pressing kick in by themselves for 20–150 ms	Above 45% heat you get random glitches, static, stuck sound and moments of silence. The HEAT LED blinks fast and the screen shakes
Rule of thumb	Hold 1 GLITCH → runaway after about 18 s; 2 → about 4 s; 3 → about 2 s	1 glitch key is mostly fine; 2 → about 6 s; 3 → about 5 s
Cooling	Let go and it settles in 2–7 s. RESET (Esc) sets heat to 0 (but stops the CPU too)	Let go and it settles in about 8 s. Esc (RESET) sets it to 0 instantly

★ Tips for riding the edge

- Tap glitches short and on the beat rather than holding them. Less heat, and it grooves.
- When the runaway starts, release just one. Heat depends a lot on how many you hold.
- The toy PC warns you with a "Too hot!" bubble. On TELEKEY, keep an eye on the HEAT LED.
- Runaways are part of the performance! If you like it, stay hot on purpose.

Real crashes: the sequencer's CRASH



During CRASH: the sound sticks, then goes silent and the screen freezes with "SYSTEM HALTED". When it ends, it reboots automatically

Put a **CRASH→reboot** note on the sequencer's ⚡ system row to crash on purpose for its length. The auto-composer puts one in the break when CHAOS is high, and always at the end in the "Collapse" style.

How to recover

- 1 **RESET**: the toy RESET button (on the toy PC, Esc or RESET / REBOOT). The CPU stops and starts again right away. When you hear "HELLO", it is back!
- 2 A sequencer CRASH resets and reboots automatically when the note ends. To stop it, press ■ (stop).

⚠ Careful!

If the toy PC LOOP switch is at the bottom (MUTE), it will not start. Move the switch up or to the middle and it starts right away.

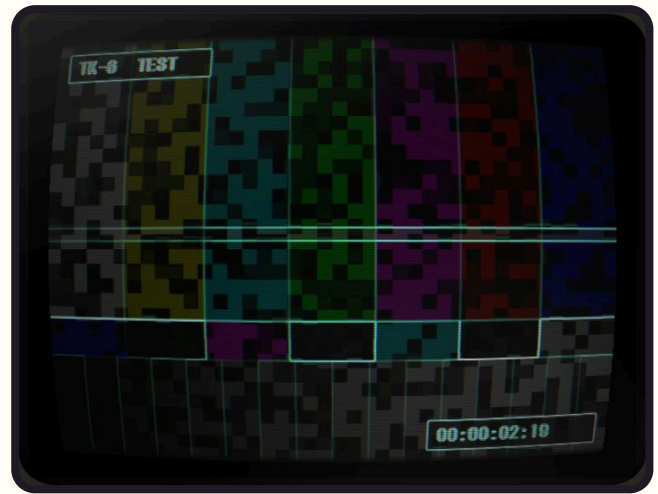


The video machine

Wreck video with TELEKEY TK-6!



Test video (normal)



Holding E (DATAMOSH) and G (EDGE)

3 kinds of input (+ test video)

Button	What you see	How to start
CAM	Webcam	The camera starts automatically the first time. Choose "Allow" when the browser asks; otherwise you get TEST
Capture a TAB	Picture and sound from another tab	Steps below. You can wreck both picture and sound
FILE	A video file of your own	Press the button and choose one, or drag & drop a video onto the screen
TEST	An automatically generated test video (colour bars, shapes, tunnel, test card) + test-signal sound	Just press the button. Free to use

Sharing a tab (Chrome)

1

Play the video you
want in
another
tab

2

Below the TELEKEY
monitor, press
"Capture
a TAB"

3

In Chrome's dialog
choose
"Tab"
and pick that tab

4

turn
on
"Also
share

5

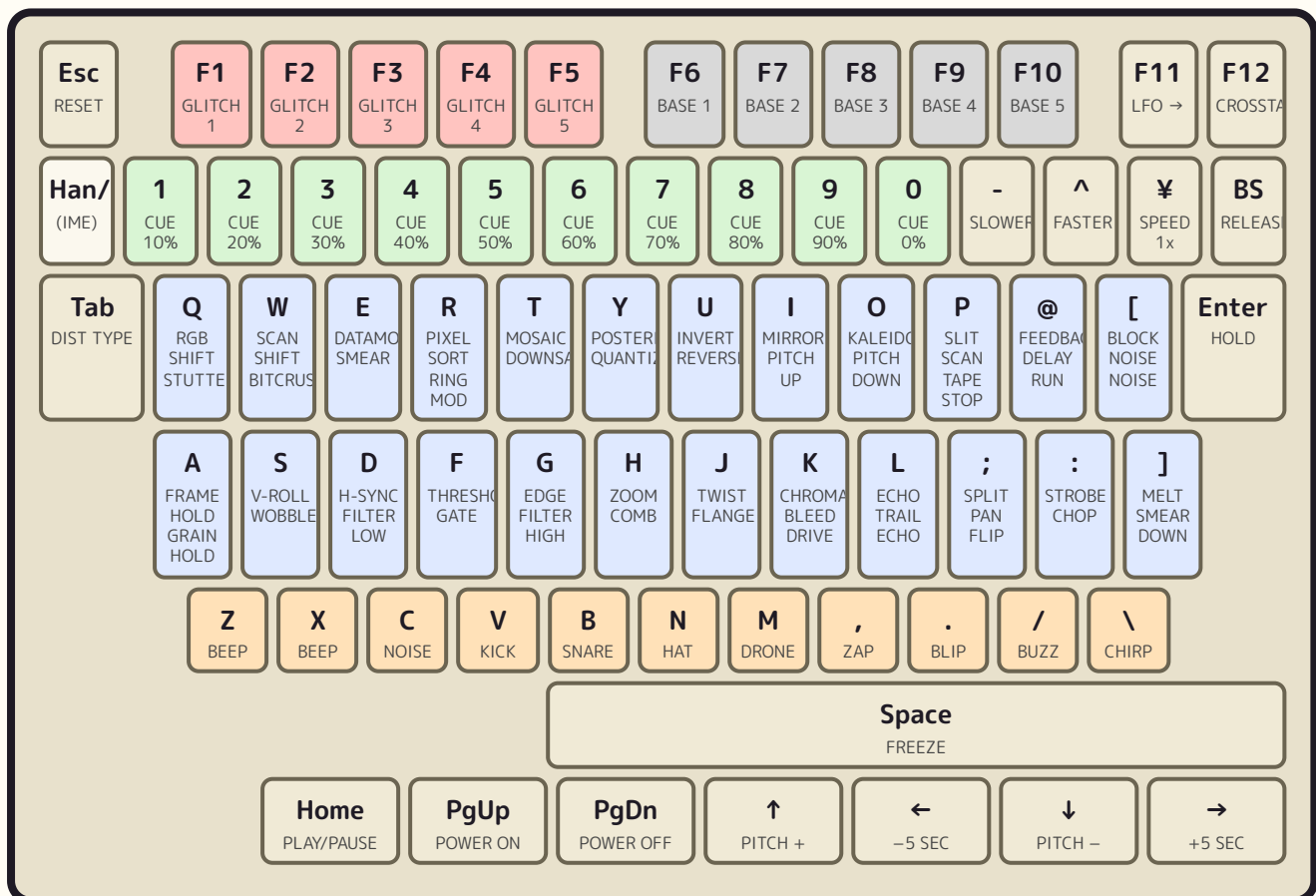
Press

! The captured tab's own sound is muted automatically, so you only hear the wrecked sound

⚠ Careful!

If "TAB (no audio...)" appears below the monitor, you skipped step 4. Capture again.
Stopping the capture goes back to TEST.

Key map



Glitch

Instruments

GLITCH buttons

BASE

Cue

Function

24 picture + sound glitches

They work while held. Press as many as you like (heat builds up accordingly).

Keys	What happens to the picture	What happens to the sound
Q	RGB SHIFT	STUTTER
W	SCAN SHIFT	BITCRUSH
E	DATAMOSH	SMEAR
R	PIXEL SORT	RING MOD
T	MOSAIC	DOWNSAMPLE
Y	POSTERIZE	QUANTIZE
U	INVERT	REVERSE
I	MIRROR	PITCH UP
O	KALEIDO	PITCH DOWN
P	SLIT SCAN	TAPE STOP
@	FEEDBACK	DELAY RUN
[	BLOCK NOISE	NOISE
A	FRAME HOLD	GRAIN HOLD
S	V-ROLL	WOBBLE
D	H-SYNC	FILTER LOW
F	THRESHOLD	GATE
G	EDGE	FILTER HIGH
H	ZOOM	COMB
J	TWIST	FLANGE
K	CHROMA BLEED	DRIVE
L	ECHO TRAIL	ECHO
;	SPLIT	PAN FLIP
:	STROBE	CHOP
]	MELT	SMEAR DOWN

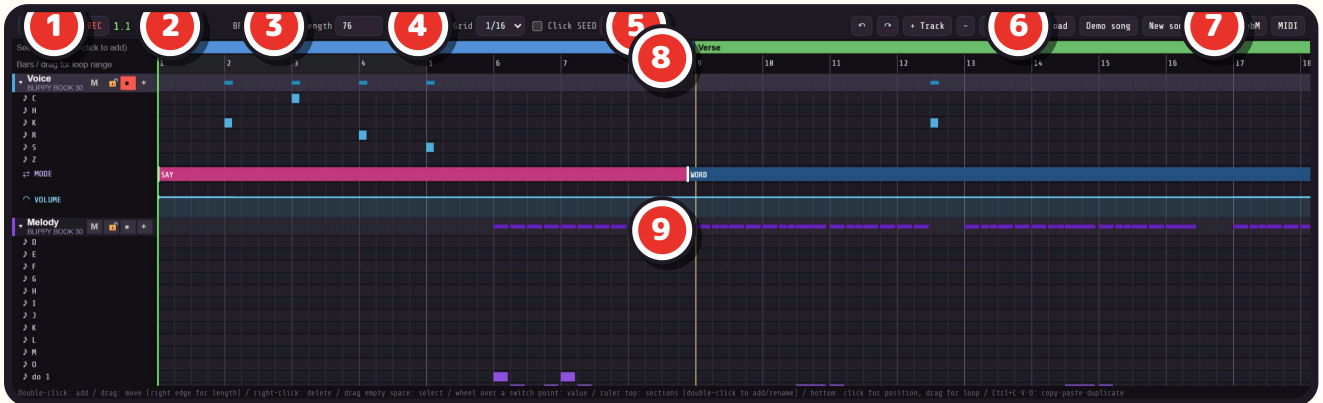
★ Record it

Use ● **REC VIDEO** at the top left of the monitor to save the wrecked picture and sound as WebM. Press again to stop and save. For sound only, use REC (WAV) in the top bar.



Sequencer & auto-composer

Record it, fix it, let it compose!



Sequencer (studio; ≡ SEQ in the rack is the same screen)

- 1** **REC**
To start, play / stop, record what you play (overdub into tracks marked ●).
- 2** **Position**
Bar, beat, time and the current section name.
- 3** **BPM / length**
Tempo and song length (bars).
- 4** **LOOP / grid / click**
Repeat the loop range, grid resolution for placing notes, metronome.
- 5** **SEED**
The song's random seed. Playing from the top resets the toys and uses this seed, so you get the same sound and glitches every time.
- 6** **Save / Load**
Save / load the song as a JSON file (it is also saved automatically in this browser).
- 7** **WAV / WebM / MIDI**
Export. WAV = sound (rendered fast), WebM = with video (takes as long as the song), MIDI = for future VST / DAW use.
- 8** **Ruler**
Top = section names (double-click to add / rename), bottom = click for playhead, drag for loop range.
- 9** **Tracks and rows**
Left: track names (M mute, 🔒 lock, ● record target, + add a row). Right: notes and knob moves.

Recording

- 1** Press ● REC (playback starts if stopped)
- 2** Play the toys. Keys, buttons and knob moves are all recorded

- 3 Each time the loop ends, it's layered into the song (overdub). Press ● REC again or ■ to finish

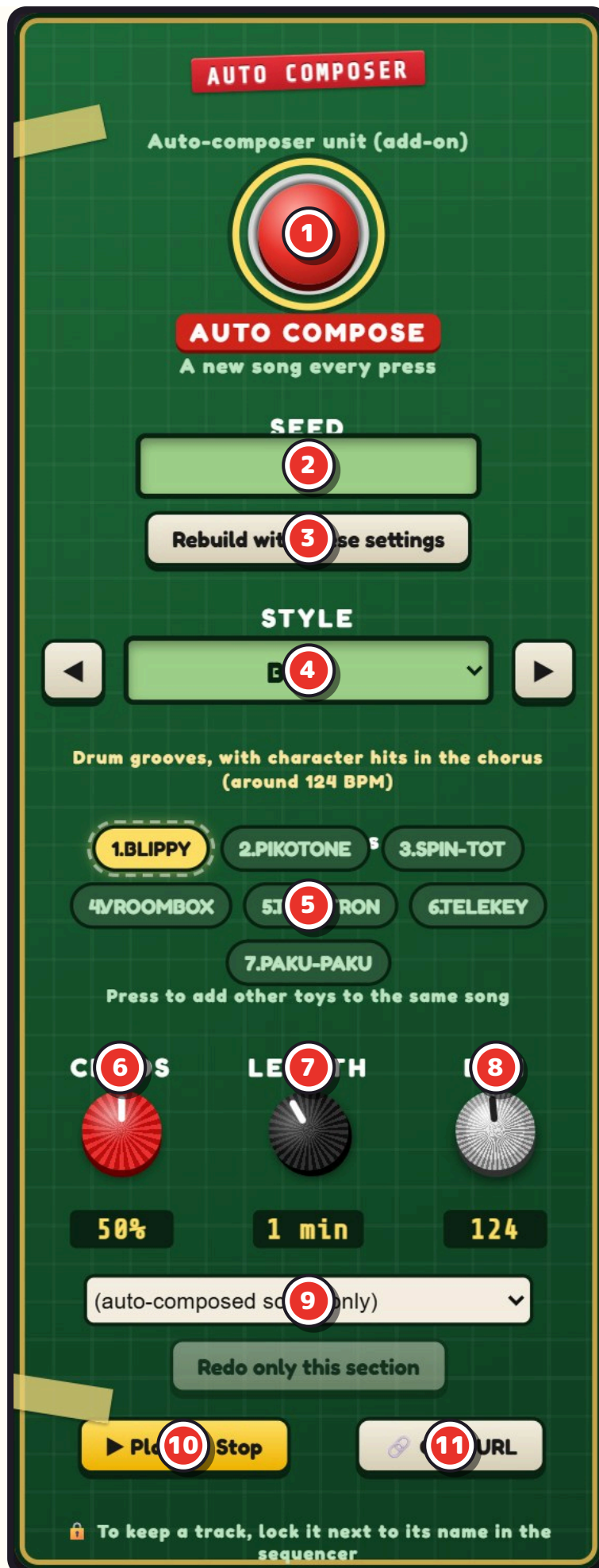
Editing

To do this	Do this
Add a note	Double-click on a row
Move / change length	Drag (drag a note's right edge for length)
Delete	Right-click, or select and press Delete
Select many	Drag in empty space
Draw knob moves	Double-click on a $\cap$ knob row to add points, drag to move them (points are joined smoothly)
Change a switch value	Mouse wheel over a point on a $\rightleftharpoons$ row
Add a row	The + to the right of the track name
Copy / paste / duplicate	Ctrl+C, Ctrl+V (at the playhead), Ctrl+D (right after)
Undo / redo	Ctrl+Z · Ctrl+Y
Section names	Double-click the top of the ruler
Loop range	Drag the bottom of the ruler

Saving and exporting

Songs save automatically in this browser. "Save" makes a JSON file, and you can export as **WAV** (sound), **WebM** (with video), **MIDI** (for future VST / DAW use).

The auto-composer unit



Auto-composer unit (AUTO COMPOSER)

1 AUTO COMPOSE

Each press writes a song with a new seed into the sequencer and plays it from the top.

3 Rebuild with these settings

Rebuilds with the same seed but the changed STYLE, CHAOS, length and BPM.

5 Joining toys

Toys you light up join the same song. One plays the beat, and melodies trade every 2 bars. The dotted toy (this panel's owner) always joins.

7 LENGTH

Song length: 30 s, 1 min, 2 min, 3 min.

9 Section select + Redo only this section

Rebuilds only the chosen section (e.g. the chorus). Nothing else changes.

11 Copy URL

Copies this song's URL. Whoever opens it gets the same song.

2 SEED

The current song's seed number. Type a number and press Enter to build that seed's song.

4 STYLE (◀ ▶)

26 styles: enka, ondo, reggae, ska, bossa nova, jazz, house, drum & bass, IDM and more. ◀ ▶ step through them, the middle opens the list. The style's description is shown below (→ table in chapter 9).

6 CHAOS

Amount of glitches, distortion and crashes (0–100%).

8 BPM

Tempo. Changing STYLE resets it to that style's usual tempo.

10 Play / Stop

Plays the song from the top / stops it.

Styles at a glance

STYLE	What it sounds like	Tempo	Glitch	Crash
Toy Plain	Toy as it is. Reads slowly and sings gently	108 (84–132)	★☆☆☆☆	None
Toy Beat	Drum grooves, with character hits in the chorus	124 (100–150)	★★★★☆	In the break at CHAOS 55% or more
Toy Ambient	Long notes float and slowly sway	80 (60–100)	★☆☆☆☆	None
Toy Noise	A storm of sound effects and glitches	132 (100–170)	★★★★★	In the break at CHAOS 35% or more

Toy Collapse	Falls apart bit by bit and always crashes at the end	116 (90–140)	★★★★★	Always (collapses at the end)
Pop Nursery	A slow children's song on the yonanuki major scale	96 (80–112)	★☆☆☆☆	None
Pop March	A sports-day march, loud and cheerful	116 (100–128)	★☆☆☆☆	None
Pop Chiptune	Old game-console style. Fast arpeggios and square waves	150 (128–172)	★★★★★	In the break at CHAOS 80% or more
Pop Punk	Fast! Loud! Eighth notes all the way	184 (160–210)	★★★★★	In the break at CHAOS 50% or more
Pop Lo-fi	A loose swung beat with dusty seventh chords	80 (68–92)	★☆☆☆☆	None
Dance House	Four-on-the-floor, off-beat hats and bouncy chords	124 (118–130)	★★★★★	In the break at CHAOS 80% or more
Dance Techno	Four-on-the-floor and 16th hats. The same sound slowly morphs	132 (124–142)	★★★★★	In the break at CHAOS 70% or more
Dance Drum & Bass	Fast 174 BPM breakbeats with a long, fat bass	174 (160–180)	★★★★★	In the break at CHAOS 60% or more
Dance Jungle	Chopped-up breaks and a growling bass	166 (155–175)	★★★★★	In the break at CHAOS 50% or more
Dance Trap	Slow heavy kicks with fast rolling hi-hats	140 (130–150)	★★★★★	In the break at CHAOS 70% or more
Dance Gabber	Distorted four-on-the-floor kicks at full speed	185 (170–200)	★★★★★	In the break at CHAOS 45% or more
World Reggae	One-drop on beat 3 and off-beat guitar (skank)	76 (66–90)	★☆☆☆☆	None

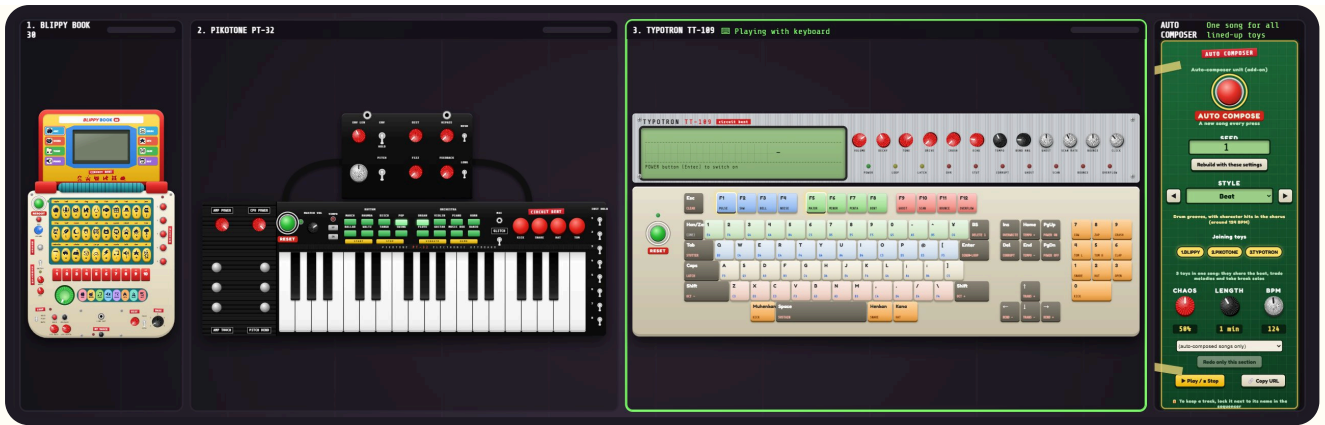
World Dub	Reggae stripped in and out, with deep echoes	72 (62–84)	★ ★ ★ ★ ★	None
World Ska	Fast off-beats, everybody jumps	164 (148–180)	★ ★ ★ ★ ★	In the break at CHAOS 90% or more
World Bossa nova	Whispering rhythm and seventh-chord guitar	128 (112–144)	★ ★ ★ ★ ★	None
World Funk	One Dorian chord with bouncing ghost notes	104 (92–118)	★ ★ ★ ★ ★	In the break at CHAOS 90% or more
World Jazz	Swing feel with a walking bass	150 (120–200)	★ ★ ★ ★ ★	None
Japanese Enka	Slow, on the yonanuki minor scale, with vocal-style turns on long notes	72 (60–84)	★ ★ ★ ★ ★	None
Japanese Ondo	Yonanuki major scale and festival drums. Dance in circles	120 (104–132)	★ ★ ★ ★ ★	None
Experimental IDM	Different math rhythms every time: chopped, shifted, rebuilt	120 (90–160)	★ ★ ★ ★ ★	In the break at CHAOS 60% or more
Experimental Breakcore	Chopped breaks shuffled apart until everything breaks	190 (170–220)	★ ★ ★ ★ ★	In the break at CHAOS 30% or more

★ Using the lock

Press  next to a track name in the sequencer to make it . A locked track **stays as it is** when you re-compose. For example: "keep the beat I like and redo only the melody". "Redo only this section" rebuilds just the chosen section (e.g. the chorus) without changing a single note elsewhere.

One song from several toys

Press the auto-composer's **"Joining toys"** to light them up and those toys join the same song (in the rack, the toy the panel belongs to always joins). Hidden toys keep playing in the rack, so you can listen while switching tabs.



Toy PC, PIKOTONE and TYPOTRON lined up with "Choose toys"

The auto-composer at the right of the studio writes one song for the lined-up toys (all of them at first). **Only one toy plays the beat**, in the verse they **trade melodies every 2 bars**, crashes happen **all at once**, and in the break **only one toy remains**.

Uncle
Mod

With "🔗 Copy URL" you can send the song's URL to a friend. Their browser rebuilds the same lineup and the same song!



The sampler

Eat, chop and arrange sounds with PAKU-PAKU 16!

A sampler made by modding a children's recording toy (with a crocodile-mouth speaker). Works in the browser on **phones, tablets and computers**. Open it with **SAMPLER** in the rack/studio top bar. The small version (toy 7 in the rack, also in the studio; chapter 3) plays, records and auto-composes like the other toys.



PAKU-PAKU 16 (landscape; in portrait the pads go to the bottom)

1 Screen

The current pad (A-01 etc.), sound name and length, waveform (red lines = START and END). The bottom row shows messages, tempo, active effects and input/output levels. On the PATTERN and SONG tabs it shows the pattern.

3 4 knobs

They do what the page says. Drag up/down, double-click to reset.

2 Knob pages

SOUND (volume, pan, pitch, fine), FILTER/ENV, SAMPLE (START, END, LOOP, velocity), MIX (mute group, bus, sound BPM), FX, SEQ, BEND.

4 Function tabs

PAD (recording, playback), PLAY (roll, 16 levels, tempo), EDIT (waveform, chop), FX, BEND, PATTERN, SONG: the buttons below change.

5 **REC / RESAMPLE**

REC → press the pad to record into → make a sound → REC to stop. RESAMPLE records what you are playing.

7 **VOL and STOP**

Master volume, and a button that stops everything (pattern playback too).

9 **16 pads**

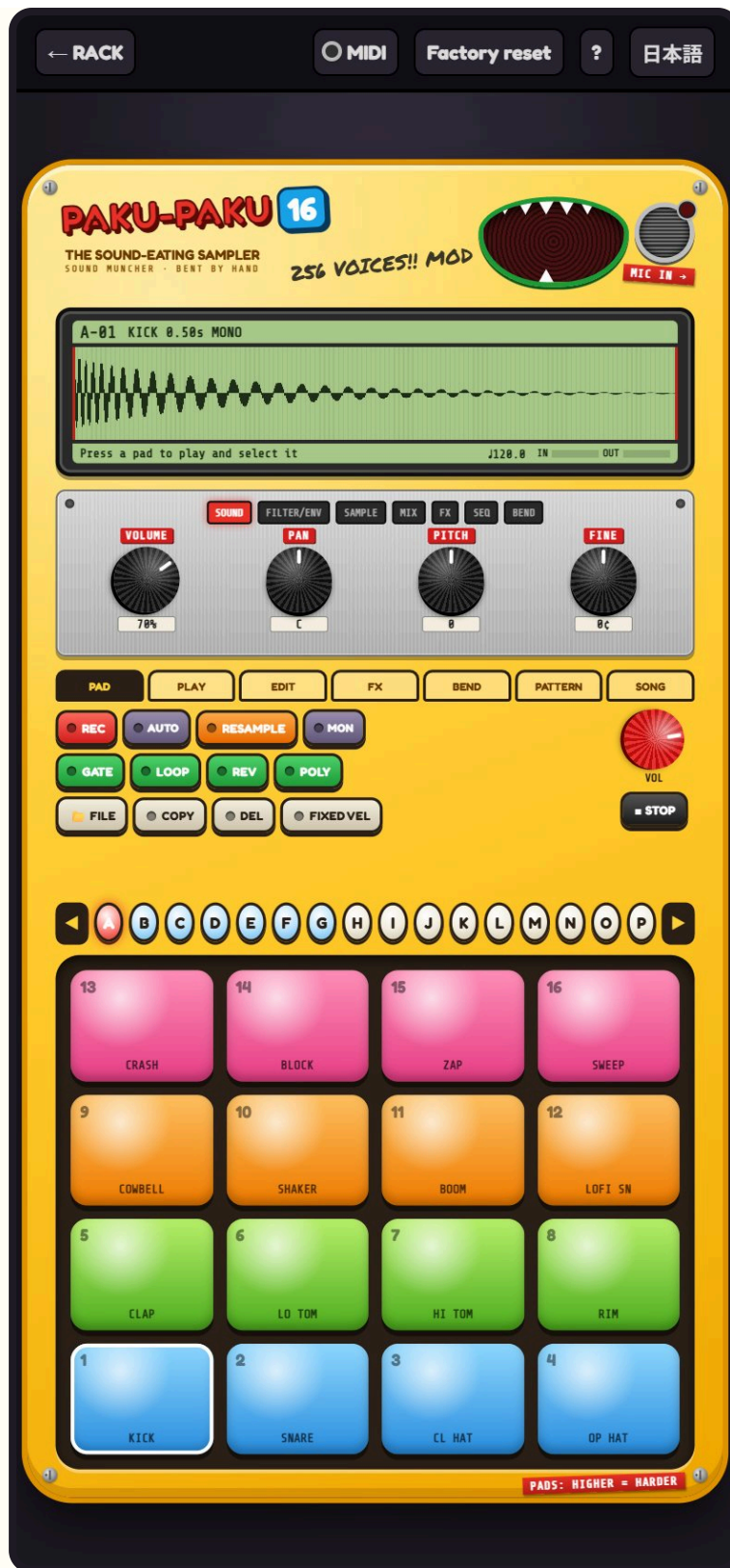
Bottom left is 1. Press to play and select it; higher = harder. In some modes they pick patterns or become steps (16th grid).

6 **GATE / LOOP / REV / POLY**

Plays only while held, loops, reverses, layers on retrigger (per pad).

8 **Banks A-P**

16 pads × 16 banks = 256 sounds. Banks with sounds are light blue.



Portrait on a phone

From zero to a song

1

Load

: REC (mic),
RESAMPLE (your
own output), 📁
FILE. 32 drum and
toy sounds come
built in

2

Chop

: ✂ CHOP on the
EDIT tab slices a
loop onto an empty
bank

3

Record

: hit pads with ●
REC on the
PATTERN tab, or
place notes on a
16th grid with
STEP

4

Chain

: on the SONG tab,
chain patterns with
+

5

Export

: WAV, MIDI, project save. → STUDIO sends it to the studio sequencer

What each tab does

Tab	What you can do
PAD	REC (AUTO: starts when sound arrives), RESAMPLE, MON (hear the input), GATE, LOOP, REV, POLY, 📁 FILE (or drop a file on a pad on PC), COPY, DEL, FIXED VEL (always full strength)
PLAY	ROLL (repeats at 1/4–1/32 or triplets while held), SUB PAD (replay the last pad), 16 LEVELS (one sound over 16 pads: pitch, strength, filter, attack, start), TEMPO and TAP
EDIT	~ WAVE EDIT (drag START / END / LOOP, 0 SNAP, ZOOM), ✂ CHOP (equal, by attacks, by hand), ⌚ TEMPO FIT (estimate BPM, then stretch keeping pitch / speed only), NORMALIZE, REVERSE, TRIM, UNDO, ↓ WAV
FX	One of 24 effects each on BUS 1, BUS 2 (chosen per pad) and MASTER (table below)
BEND	6 jumper wires on a bare circuit board. Heat builds into a runaway (unplug to cool)
PATTERN	▶ PLAY, ● REC (overdub, aligned by QUANT), METRO, ERASE, ⌂, PATTERN (choose P01–P16 with the pads), 1–8 bars, CLEAR, STEP, SWING (knob)
SONG	Chain patterns, repeats, ▶ SONG, ↓ WAV (pattern / song), ↓ MIDI, 💾 save / 📁 load (.paku), → STUDIO



A beat placed with STEP (screen: across = time, rows = pads; red line = playhead)



✂ CHOP: the bank B loop sliced automatically by its attacks (yellow lines are the cuts)

24 effects

#

Effect

Knobs 1-4

1	FILTER+DRIVE	CUTOFF · RESO · DRIVE · TYPE
2	ISOLATOR	LOW · MID · HIGH · MIX
3	LO-FI	BITS · RATE · TONE · MIX
4	VINYL SIM	NOISE · WOW · AGE · MIX
5	CASSETTE SIM	WOW · SATURATE · HISS · TONE
6	SYNC DELAY	TIME · FEEDBACK · TONE · MIX
7	TAPE ECHO	TIME · FEEDBACK · WOBBLE · MIX
8	REVERB	SIZE · DECAY · TONE · MIX
9	CHORUS	RATE · DEPTH · WIDTH · MIX
10	FLANGER	RATE · DEPTH · FEEDBACK · MIX
11	PHASER	RATE · DEPTH · FEEDBACK · MIX
12	COMPRESSOR	THRESH · RATIO · RELEASE · MAKEUP
13	DJ LOOPER	LENGTH · HOLD · SPEED · REVERSE
14	SLICER	RATE · PATTERN · DEPTH · SMOOTH
15	RING MOD	FREQ · WOBBLE · TONE · MIX
16	TREMOLO/PAN	RATE · DEPTH · MODE · SHAPE
17	AUTO WAH	SENS · LFO · RESO · MIX
18	EQ	LOW · MID · HIGH · MID FREQ
19	DISTORTION	DRIVE · TONE · LEVEL · MIX
20	PITCH SHIFT	PITCH · WINDOW · FEEDBACK · MIX
21	TAPE STOP	TIME · STOP · MIX
22	SCATTER	RATE · DEPTH · TYPE · MIX
23	RESONATOR	NOTE · FEEDBACK · TONE · MIX
24	TOY SPEAKER	SIZE · RATTLE · CRUNCH · MIX

Circuit bending (BEND tab)



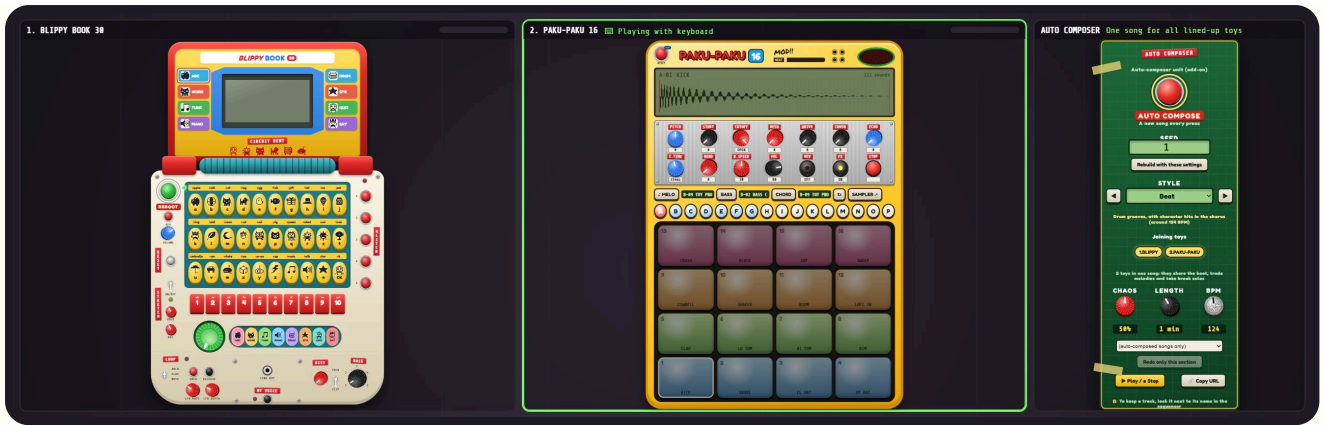
CLOCK, BIT ROT and SAG wires plugged in

Wire	What happens
CLOCK	Broken clock: playback speed jumps around
STUCK	Stuck buffer: repeats a tiny moment
RUNAWAY	Runaway pitch: drifts up and down, then snaps back
BIT ROT	Rotting bits: crunch and crackle
CROSSTALK	Crosstalk: neighbouring pads sound quietly too
SAG	Dying battery: loud parts sag, get muffled and drop in pitch

★ Bending tips

The **AMOUNT** knob sets the strength, and **SPEED** how often it happens. The more wires and the louder the sound, the more **HEAT** builds up, and when the body glows red you get a **Runaway** (all wires briefly hit harder). Unplug to cool down. WAV exports break in exactly the same way.

Using it with the studio (DAW)



BLIPPY BOOK 30 and PAKU-PAKU 16 lined up in the studio

Toy 7 in the rack, and in the studio under "Choose toys": **PAKU-PAKU 16 (sampler)**. It has an auto-composer too, playing factory instruments that suit the style (taiko, bamboo flute and shamisen for ondo, skank guitar and organ for reggae, ride and sax for jazz...) on drums and the ♪ MELO, BASS and CHORD pads for melody, bass and chords (it can join group songs with other toys). Sounds you made on the sampler page (saved in this browser) play as they are, and in the sequencer they are rows A-01 to P-16. Press → **STUDIO** on the sampler's SONG tab to put your song into the studio sequencer and open it. After changing sounds, press "⌂" (reload) on the studio sampler.

⚠ Recording with a microphone

The browser asks whether the page may use the microphone: **allow it**. On a phone, open the public (https) page. Turning on MON while sound comes out of the speakers can cause feedback (headphones recommended).

Uncle
Mod

On a PC, Z X C V · A S D F · Q W E R · 1 2 3 4 are laid out just like the pads.
MIDI pad controllers play them with notes 36–51!



Recommended seeds

Listen to these seeds!

Picked from auto-composed songs with clear build-ups and little clipping. Open a link and the same song is built (press "▶" to play).

Studio (toy PC + TELEKEY together, 1 min)

STYLE	SEED	CHAOS	Build-up	Link
Plain	712970	30%	★★★☆☆	Hear in the studio ▶
Plain	819865	30%	★★★☆☆	Hear in the studio ▶
Beat	813925	60%	★★★★☆	Hear in the studio ▶
Beat	306848	60%	★★★★☆	Hear in the studio ▶
Ambient	634954	35%	★★★★☆	Hear in the studio ▶
Ambient	768829	35%	★★★★☆	Hear in the studio ▶
Noise	479799	80%	★★★★☆	Hear in the studio ▶
Noise	472087	80%	★★★★☆	Hear in the studio ▶
Collapse	392221	80%	★★★★☆	Hear in the studio ▶
Collapse	402280	80%	★★★☆☆	Hear in the studio ▶
Nursery	953085	50%	★★★☆☆	Hear in the studio ▶
Nursery	260012	50%	★★★☆☆	Hear in the studio ▶
March	224234	50%	★★★☆☆	Hear in the studio ▶
March	759961	50%	★★★☆☆	Hear in the studio ▶
Chiptune	13970	50%	★★★☆☆	Hear in the studio ▶
Chiptune	255520	50%	★★★☆☆	Hear in the studio ▶
Punk	461303	50%	★★★☆☆	Hear in the studio ▶
Punk	306746	50%	★★★☆☆	Hear in the studio ▶
Lo-fi	35278	50%	★★★★☆	Hear in the studio ▶
Lo-fi	308232	50%	★★★★☆	Hear in the studio ▶
House	580772	50%	★★★★☆	Hear in the studio ▶

House	307569	50%	★★★★★	Hear in the studio ▶
Techno	709344	50%	★★★★★	Hear in the studio ▶
Techno	504383	50%	★★★★★	Hear in the studio ▶
Drum & Bass	473028	50%	★★★★★	Hear in the studio ▶
Drum & Bass	899622	50%	★★★★★	Hear in the studio ▶
Jungle	684952	50%	★★★★★	Hear in the studio ▶
Jungle	73187	50%	★★★★★	Hear in the studio ▶
Trap	209518	50%	★★★★★	Hear in the studio ▶
Trap	393274	50%	★★★★★	Hear in the studio ▶
Gabber	739488	50%	★★★★★	Hear in the studio ▶
Gabber	432894	50%	★★★★★	Hear in the studio ▶
Reggae	158293	50%	★★★★★	Hear in the studio ▶
Reggae	928075	50%	★★★★★	Hear in the studio ▶
Dub	859821	50%	★★★★★	Hear in the studio ▶
Dub	187943	50%	★★★★★	Hear in the studio ▶
Ska	688045	50%	★★★★★	Hear in the studio ▶
Ska	274836	50%	★★★★★	Hear in the studio ▶
Bossa nova	163022	50%	★★★★★	Hear in the studio ▶
Bossa nova	900145	50%	★★★★★	Hear in the studio ▶
Funk	804134	50%	★★★★★	Hear in the studio ▶
Funk	892426	50%	★★★★★	Hear in the studio ▶
Jazz	378585	50%	★★★★★	Hear in the studio ▶
Jazz	258794	50%	★★★★★	Hear in the studio ▶
Enka	210164	50%	★★★★★	Hear in the studio ▶
Enka	380847	50%	★★★★★	Hear in the studio ▶
Ondo	626218	50%	★★★★★	Hear in the studio ▶
Ondo	778305	50%	★★★★★	Hear in the studio ▶
IDM	918858	50%	★★★★★	Hear in the studio ▶
IDM	45011	50%	★★★★★	Hear in the studio ▶

Breakcore	424259	50%	★★★★★	Hear in the studio ►
Breakcore	22289	50%	★★★★★	Hear in the studio ►

Toy PC solo (1 min)

STYLE	SEED	CHAOS	Build-up	Link
Plain	893490	30%	★★★★★	Hear in the rack ►
Beat	257361	60%	★★★★★	Hear in the rack ►
Ambient	246540	35%	★★★★★	Hear in the rack ►
Noise	141323	80%	★★★★★	Hear in the rack ►
Collapse	866949	80%	★★★★★	Hear in the rack ►
Nursery	557647	50%	★★★★★	Hear in the rack ►
March	899268	50%	★★★★★	Hear in the rack ►
Chiptune	62048	50%	★★★★★	Hear in the rack ►
Punk	401747	50%	★★★★★	Hear in the rack ►
Lo-fi	583170	50%	★★★★★	Hear in the rack ►
House	501122	50%	★★★★★	Hear in the rack ►
Techno	913292	50%	★★★★★	Hear in the rack ►
Drum & Bass	255759	50%	★★★★★	Hear in the rack ►
Jungle	26301	50%	★★★★★	Hear in the rack ►
Trap	759121	50%	★★★★★	Hear in the rack ►
Gabber	567720	50%	★★★★★	Hear in the rack ►
Reggae	192208	50%	★★★★★	Hear in the rack ►
Dub	275773	50%	★★★★★	Hear in the rack ►
Ska	319483	50%	★★★★★	Hear in the rack ►
Bossa nova	653406	50%	★★★★★	Hear in the rack ►
Funk	694490	50%	★★★★★	Hear in the rack ►
Jazz	412106	50%	★★★★★	Hear in the rack ►
Enka	473540	50%	★★★★★	Hear in the rack ►

Ondo	953460	50%	★ ★ ★ ★ ★	Hear in the rack ▶
IDM	79682	50%	★ ★ ★ ★ ★	Hear in the rack ▶
Breakcore	838967	50%	★ ★ ★ ★ ★	Hear in the rack ▶

"Build-up" roughly shows the volume difference between the quietest and the busiest section. How they were chosen: 4 candidates per style were actually rendered and scored on that difference, a comfortable level and little clipping.



Tricks and secrets

Things to brag about!

★ SPIN-TOT's hidden rhythms 22–28

Press rhythm key 4 for the hidden patterns 22–28: odd meters of 5, 7, 11, 13, 9, 15 and 6 steps, with scattered sounds too.

★ The toy PC's screen mutates

During BASE 5 (VOICE) MISREAD, SCRAMBLE and GARBAGE, not just the voice but also the screen's letter and character switch to the misread key.

★ It won't switch on!?

If the toy PC LOOP switch is at the bottom (MUTE), it will not start, just like the real thing. A speech bubble tells you; move it up or to the middle (it starts right away).

★ Play the engine with the radio

With VROOMBOX's PRESET HIJACK on, radio presets 1–8 become an engine scale (C to C).

★ Storing cue points

On TELEKEY, Shift+number stores the current position on that number key.

★ Overwriting the line loop

On TYPOTRON, turn on Ins (overwrite) during an Enter line loop and type: the note playing now gets replaced.

★ Holding down the instrument lines

PIKOTONE's INST HOLD is a mod that holds down the instrument-select lines. The waveforms of the sounds you turn on blend together.

★ The same glitches every time

Playing a song from the top in the sequencer resets the toys and uses the seed, so even the glitches are identical every time. Runaways happen in the same places!

★ TELEKEY's startup sound

When it starts, the CRT goes "boom" and "beep". It also plays when rebooting after RESET or a crash.



Troubleshooting (Q&A)

Don't panic, read this

Q. No sound!

A. Browsers stay silent until you first touch the page, so touch anywhere first (that also switches the power on). Also check the volume knob, your computer volume, and that the browser tab is not muted. The toy PC is silent when the LOOP switch is at MUTE (bottom).

Q. Keys don't respond

A. Keys may not respond while it is starting (during the startup sound). In the studio the keyboard goes to the toy you clicked (green frame). After clicking the sequencer the keyboard belongs to the sequencer, so click the toy again. The toy PC ignores keys until its startup "HELLO" ends.

Q. The F keys don't work

A. Browsers sometimes grab F6, F7, F11, F12 and so on first. If so, click the on-screen keys with the mouse.

Q. No picture (TELEKEY)

A. If you did not allow the camera, it shows TEST (test picture). If TAB says "no audio", you forgot to turn on "Also share tab audio" in the share dialog.

Q. The screen is all white / black

A. TELEKEY uses WebGL2. If the monitor says "WebGL2 is not available", check that "hardware acceleration" is on in your browser settings or try a recent Chrome. Reloading the page (F5) helps too.

Q. It won't come back from a crash

A. Even if RESET stops the CPU, it starts again right away. A sequencer CRASH automatically does RESET → reboot when its note ends. If it will not stop, press ■ (stop) to get back. If it is only running wild from heat, let go and it cools down.

Q. The same URL gives a different song

A. When the auto-composer program is updated, the same settings can give a different song. On the same version of the page it is always the same.

Q. Where are my songs saved?

A. Automatically inside this browser (not on other computers or browsers). "Save" in the sequencer makes a JSON file; "Load" brings it back.

Q. Does it work on phones and tablets?

A. It opens, though small. But this app is built around playing with a PC keyboard, so Chrome on a PC is the best choice. (The PAKU-PAKU 16 sampler page is designed for phones.)



Glossary

Hard words in one line

circuit bending	Deliberately rewiring toys and electronics or adding parts to get sounds their designers never imagined. This app imitates that inside your computer.
Heavy mod	Adding lots of buttons and knobs through circuit bending.
Glitch	Sound or picture breakups that look like a "bug". Here you cause them on purpose for fun.
BASE	The knob that switches glitch families. The same button gives a different glitch when BASE changes.
LFO	A slowly swinging wave (Low Frequency Oscillator). It wobbles pitch or strength automatically.
Loop / LOOP HOLD	Grabbing part of a sound and repeating it.
Feedback	Feeding the output (sound or picture) back in. The source of echoes, howling and runaways.
Distortion (DIST)	Deliberately distorting the sound.
Bit crush (CRUSH)	Lowering the resolution (bits and rate) for a gritty old-video-game sound.
Voltage starve	Deliberately starving it of power. The sound gets slow, low and wobbly, and finally stops.
Heat (stress)	An invisible value that rises when you overdo it. Too high and glitches run wild by themselves (runaway). Let go to cool down.
Crash	During a sequencer CRASH the sound sticks and the screen freezes. Afterwards it RESETs and reboots.
Sequencer	Records your playing (which keys when, how the knobs moved) and plays it back automatically.
Overdub	Recording more on top of what you already recorded.

Automation	Recording knob moves and replaying them. In the sequencer: the $\cap$ knob rows.
Section	A part of a song (intro, verse, chorus...).
BPM	Beats per minute: the speed of a song.
Seed	The "seed" of the random numbers. The same seed gives the same random sequence, so you get the same song and glitches.
Motif	A short phrase repeated in a song. The auto-composer repeats a 2-bar motif with small changes.
Cue points	A fixed point in a video. Press a key to jump there.
CRT	An old TV screen. TELEKEY's screen is made to look rounded and blurry like one.
MIDI	The standard for connecting electronic instruments. Keyboards and sequencers send each other data.
WAV / WebM	WAV is an audio file, WebM is a video + audio file.



About rights

This is very important

TELEKEY TK-6 can capture, process and record video and sound from tab sharing, video files and webcams.

If you process, record or publish other people's works (videos, music, images, etc.), always get permission from the person who holds the rights. Even if it was just for fun, publishing a recording on the internet can infringe the rights holder's rights.

When publishing video that shows people (including camera video), get their permission too.

The **test video and test signal** built into this app are generated by the program and free to use. Sounds you make by playing the toys themselves (keys, glitches, auto-composer) are free to use as well.

English: If you process, record, or publish someone else's copyrighted work (videos, music, images, etc.), please obtain permission from the rights holder. The built-in test pattern and test signal are procedurally generated and free to use.